Date Submitted: Not Submitted

Project Title
Verification of Sequential Chemical-Mechanical-Enzymatic Pretreatment of Abaca and Pineapple Fibers

Project Description
Studies to continuously improve our existing processes for the utilization of natural fibers and conversion into textile products are prompt by the increasing interest for natural fiber-based textile materials. These are also brought about by the need to further enhance the properties and performance of the natural textile substrates, alongside with the desire to reduce processing cost, and ultimately, come up with more competitive product cost. In 2016 the efficiency of incorporating the enzymatic process during fiber pretreatment, in addition to the chemical and mechanical treatment was studied. The said modified pretreatment process was employed to improve the processability of the resultant treated fibers and the final property of spun yarns and fabrics. Enzymatic treatment was introduced as a pretreatment step to reduce further non-cellulosic components and render the fibers softer prior to opening and blending with cotton. From initial observation, treated and opened pineapple fibers using the modifed process were softer to touch and produced smoother yarns. This observation will be evaluated against the physical and fiber component test result of the yarns submitted to the Technical Services Division (TSD). The mechanical test result will be a critical in evaluating the result of the modified pretreatment method. The enzyme can partially but minimally degrades cellulose to remove short chains organic polymers resulting to improved material handle. Thus, finding optimum enzyme concentration was one of the parameters which was evaluated during the study. In 2017, the optimized process identified in 2016 will be verified in bench scale. 2.5-3 kilos of pineapple and abaca fibers will be subjected to the modified pretreatment method. Properties of fibers will be evaluated against the result of the laboratory trials and noting significant discrepancies in values. The treated fibers will be cut, opened and blended with cotton prior to spinning into yarns. Yarns will be converted to a fabric material enough for testing to evaluate the effect of the pretreatment used in the macro-properties of fabrics. Yarns and fabrics will also be tested for its dyeing performance using both synthetic and chemical pigments.

Project Objective (summary)
General Objective To verify the sequential chemical-mechanical-enzymatic pretreatment process for abaca and pineapple fibers in bench scale Specific Objectives 1. To produce prototype yans and fabrics from the modified pretreatment process 2. To evaluate the performance/property of developed yarns and fabrics 3. To determine dyeability of the developed yarns and fabrics made from the modified pretreatment method

Project Duration	Jan 01, 2017 to Dec 31, 2017
Project Status	Completed
Approved Budget	0.00
Project Leader	Jeannie Lynn C. Perez
Co-Project Leader	N/A
Members	
Date of Preparation	Sep 06, 2019

Project Significance

Project Objectives

Project Outputs & Performance Indicators	
Expected Output (Deliverable)	**Performance Indicators**
Prototype abaca and pineapple fiber-based yarns	Prototype abaca and pineapple fiber-based yarns
Prototype abaca and pineapple fiber-based fabrics	Prototype abaca and pineapple fiber-based fabrics
Technical Paper	Technical Paper

Year #	Funding Source Name	PS		MOOE		CO	TOTAL
		Direct Cost	Indirect Cost	Direct Cost	Indirect Cost		
Total Cost		-	-	-	-	-	-

PS Breakdown						
Budget Item (LIB)		Amount				TOTAL
Staff Name	% Involvement	**Q1**	**Q2**	**Q3**	**Q4**	
Total Cost		**0.00**	**0.00**	**0.00**	**0.00**	**0.00**

MOOE Breakdown					
Budget Item (LIB)	Amount				TOTAL
	Q1	**Q2**	**Q3**	**Q4**	
Total Cost	**0**	**0**	**0**	**0**	**0**

Work Plan

Completion by Quarter

#	Major Activity	Milestone Indicator	Date Completion

Risk Management Plan		
Key Success Factor	**Risk**	**Countermeasure**

Change Control Plan

Quality Management Plan

Management & Organization

Manpower

Training & Development